Unlike conventional X-ray examinations and computed tomography (CT) scans, MRI does not utilize ionizing radiation. MRI uses a powerful magnetic field, radio frequency pulses and a computer to produce detailed pictures of organs, soft tissues, bone and virtually all other internal body structures. The radio frequency pulses re-align hydrogen atoms that naturally exist within the body while the patient is in the scanner without causing any chemical changes in the tissues. As the hydrogen atoms return to their usual alignment, they emit different amounts of energy that vary according to the type of body tissue from which they come. The MR scanner captures this energy and creates a picture of the tissues scanned based on this information (Figure12.9).

The magnetic field is produced by passing an electric current through wire coils in most MRI units. Other coils, located in the machine and in some cases, are placed around the part of the body being imaged, send and receive radio waves, producing signals that are detected by the coils. The electric current does not come in contact with the patient.

A computer then processes the signals and generates a series of images, each of which shows a thin slice of the body. The images can then be studied from different angles and interpreted by a radiologist.

Frequently, the differentiation of abnormal (diseased) tissue from normal tissues is better with MRI than with other imaging modalities such as X-ray, CT and ultrasound. Detailed MR images allow physicians to evaluate various parts of the body and determine the presence of certain diseases. The images can then be examined on a computer monitor, transmitted electronically, printed or copied to a CD or uploaded to a digital cloud server.

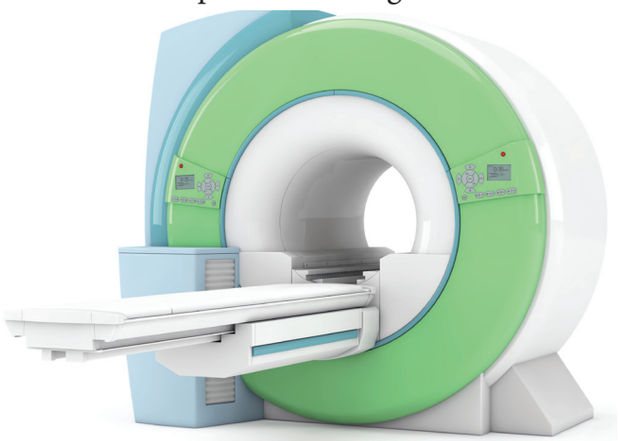


Figure 12.9 Artificially coloured MRI scan of human brain

Clinical significance

MR imaging of the body is performed to evaluate organs of the chest and abdomen, pelvic organs including the bladder and the reproductive organs, blood vessels and lymph nodes.

Physicians use an MR examination to help diagnose or monitor treatment for conditions such as:

- Tumours of the chest, abdomen or pelvis.
- Diseases of the liver, inflammatory bowel disease, heart problems, such as congenital heart disease.
- Malformation of the blood vessels and inflammation of the vessels (vasculitis).
- A foetus in the womb of a pregnant woman.
- Visualising injuries, torn ligaments especially in areas like wrist ankle or knee.

Why the magnetic resonance imaging technique gives excellent job of showing soft tissues and blood vessels?

XI Std Zoology Chapter-12 EM.indd 229 30/12/2022 14:03:42